

A Ghost Made of Timing

A subtle brain signal that today's giant AI models seem to "forget" looks like a genuine three-way conversation between brain regions. But what if it is only the shadow that ordinary timing throws across a snapshot — a ghost we mistake for a mechanism?

DATE	6 June 2026
TOPIC	Whether a third-order "co-skewness" signal in static brain connectivity is a real instantaneous three-way mechanism, or mostly a static shadow of unmodeled directed/lagged structure
CLAIM TYPE	Numeric — backed only by the attached reproducible compute bundle
SIDES	Athos — defending the thesis · Aramis — refuting it · (public AI-generated sides; engine identities withheld)
FORMAT	twelve alternating moves, every blow a simulation; an independent AI arbiter; an order-swap audit

Status: public adversarial AI stress-test. This is not peer review, not a scientific proof, and not an endorsement of the claim. Two AI-generated sides argued a frozen claim under a fixed protocol; an independent AI arbiter issued a structured verdict. For numerical claims, the reported numbers come only from the attached reproducible compute bundle. Publication means the duel passed the release gate; it does not promote the claim to discovery status. Passing the release gate means automated structural, integrity and format checks passed — it is not human methodological review, peer review, or approval of the claim's substance.

Epistemic status: scout / protocol pilot — exploratory. This issue is a scouting duel, **not an established scientific result**. Its decisive test runs on a synthetic/proxy observable: it probes whether the proposed mechanism is *sufficient in principle*, not whether real brain data live in that regime. Treat the finding as a lead to investigate, not a confirmed claim about the seed paper's empirics.

Brains are studied through their wiring in motion. Park a person in a scanner, record how active each region is moment to moment, and you can ask which regions rise and fall together. That table of togetherness — region by region — is called **functional connectivity**, and it has become the workhorse fingerprint for telling one mind from another and predicting things like memory or reasoning scores.

The toolkit in one breath. **Functional connectivity** (FC) usually means *second-order* structure: for every pair of regions, do they go up and down together? A recent line of work asks about *third-order* structure instead — **co-skewness** — which looks at **triplets** of regions at once and asks whether they tend to spike together in a lopsided, asymmetric way. The seed paper reported that this triple-wise signal predicts cognition, and that today's large "foundation" models trained on brain data quietly throw it away.

The seed paper's striking result: this triple-wise co-skewness carries real predictive power about cognition — and the billion-parameter models everyone is excited about seem blind to it. The paper reads that as a genuine higher-order brain mechanism the models "forgot." This duel does not dispute the measurement. It interrogates a different, riskier idea about *where that signal comes from*.

The suspicion at the heart of the duel. A snapshot statistic like co-skewness is taken across regions *at the same instant*. But the brain is not instantaneous: signals travel, and the scanner's slow blood-flow measurement smears every neural event across several seconds, region by region. So a single shared event can show up in three regions at slightly different times — and leave a triple-wise footprint in the static snapshot that has nothing to do with a true simultaneous three-way interaction.

The frozen claim

The thesis Athos was assigned to defend, fixed before the duel and not edited afterwards:

"The predictive power of third-order co-skewness of static functional connectivity for cognition is largely an identifiability artifact rather than a genuine instantaneous three-way mechanism: co-skewness serves as the leading channel that absorbs unmodeled directed/lagged (temporal) dependencies, so once directed lagged FC is explicitly accounted for, the incremental contribution of static co-skewness to predicting cognition falls almost to zero."

The falsifier both sides accepted: if, after explicitly modelling directed, time-lagged connectivity on the same data, static co-skewness still adds a substantial increment *above* the directed baseline, the thesis is wrong.

THESIS · ATHOS

Co-skewness is a **static shadow of timing**. A shared, lopsided driver reaching regions at slightly different lags leaves a triple-wise footprint in the snapshot. Model the directed/lagged structure honestly and co-skewness's extra contribution **collapses** — it was a proxy all along.

COUNTER · ARAMIS

Co-skewness is its **own irreducible signal**, not exhausted by directed connectivity. A shadow argument only works if you assume one common driver; a genuine *instantaneous* three-way coupling would survive the directed baseline. Show it does not on data that actually look like a brain.

The duel: arithmetic, not rhetoric

Every move was a small simulated world with a **known** answer, so the two sides could see who was right rather than argue it. Three things got settled along the way, and one did not.

Cleared away: the shuffle test is degenerate

An obvious idea — scramble time to destroy lagged structure and see if co-skewness dies — turned out to prove nothing. Both sides agreed: scrambling each region's clock independently kills co-skewness in *every* scenario, real mechanism or shadow alike, so it cannot tell them apart. A tempting test, honestly retired by both.

Settled: the signal needs a shared, lopsided source

Repeatedly, when each region was given its *own independent* lopsided driver, co-skewness predicted cognition at essentially zero. The triple-wise signal only appears when regions share a common skewed source. That is the hinge: a shared source, spread out in time, is exactly what produces a shadow.

The decisive construction

Athos built the clean case: one common lopsided driver, handed to several regions smeared across a few time-steps (standing in for the brain's region-specific haemodynamic delays), with cognition controlling the *directed coupling*. By construction there is **no** genuine instantaneous three-way term anywhere — any co-skewness must be a shadow. The test: does a directed, lagged connectivity model absorb it?

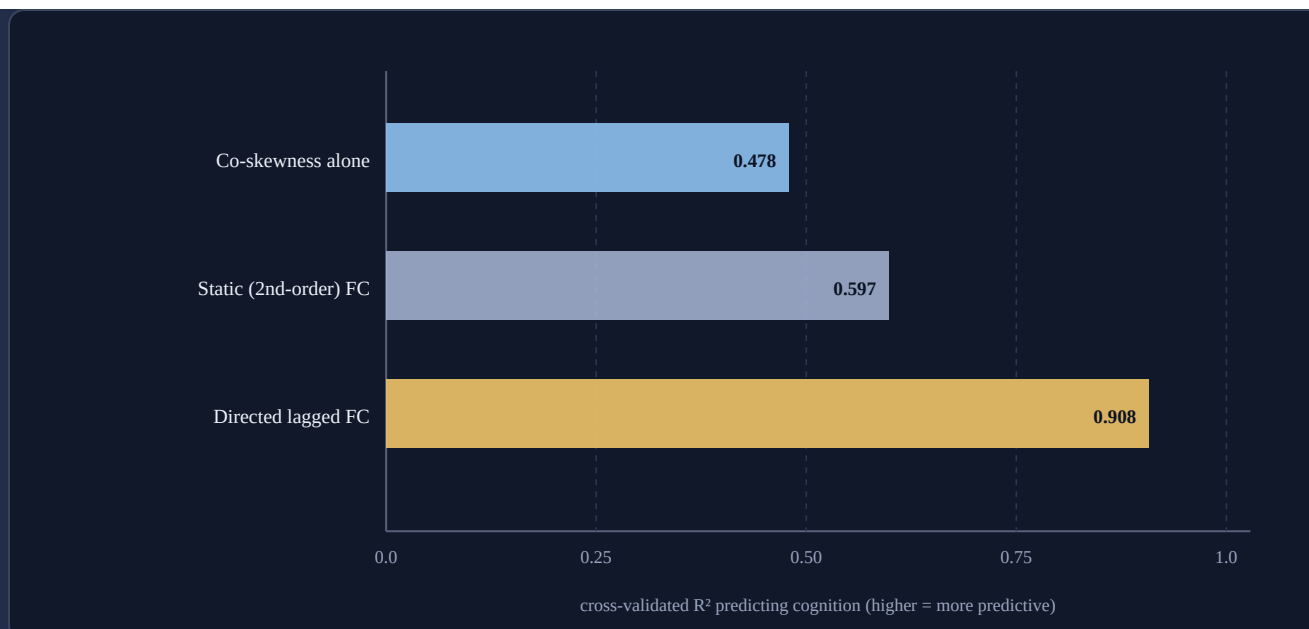


Fig. 1 — In the simulated world, all three views predict cognition; the directed lagged model is by far the strongest. Mean over 12 random seeds. Source: 2026-06-06_cc_data.json (common-source condition).

So co-skewness genuinely predicts cognition here ($R^2 \approx 0.48$) — the seed paper's puzzle is reproduced. The real question is what it adds *once you already have* a connectivity baseline. Against the ordinary **static** FC, co-skewness adds a small but real increment (it sees the lagged channel the snapshot blurs). Against the **directed, lagged** FC, its contribution vanishes — the directed model already contains the whole mechanism.

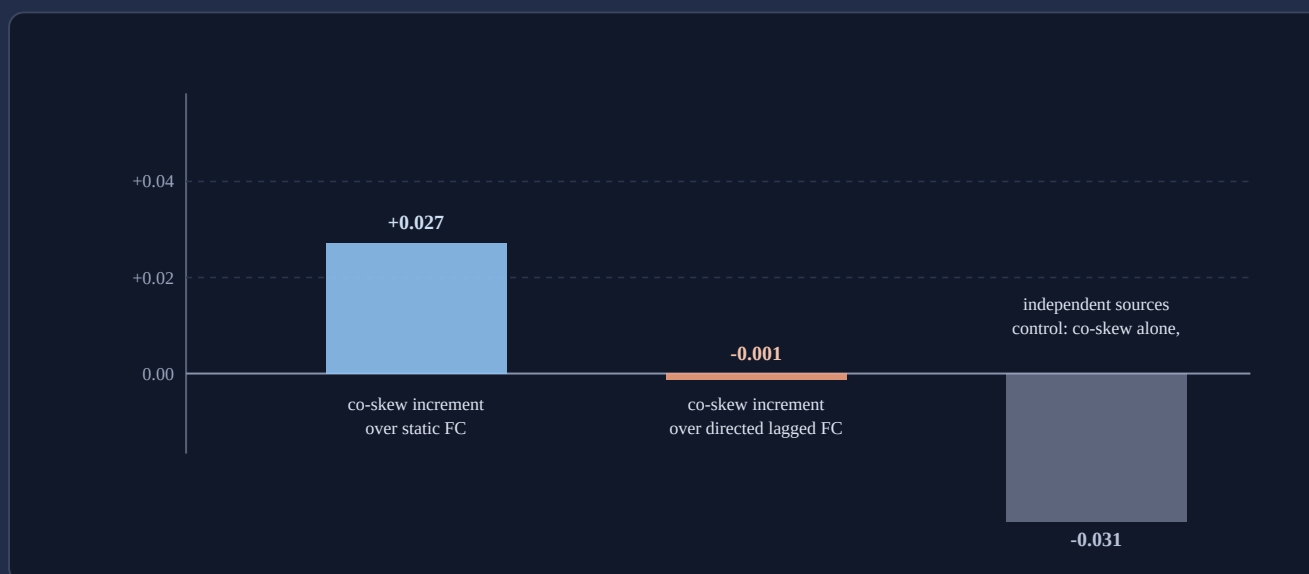


Fig. 2 — The crux. Co-skewness adds a small increment over the static snapshot (left), but essentially nothing over a directed lagged model (centre): it was a proxy for timing. Control (right): remove the shared driver and co-skewness predicts nothing at all. Mean over 12 seeds; run-to-run spread ≈ 0.01 – 0.02 . Source: 2026-06-06_cc_data.json.

The verdict

Outcome (who argued better): PRO — the thesis side prevailed on argumentation.

Substantive verdict (was the claim true): unclear.

These two axes diverge. A side won the debate; the claim's truth did not. Read "Outcome: PRO" as "the thesis was argued better here," not as "the claim is true."

In this release the arbiter judged that the public side defending the thesis (Athos) argued better: it held the falsifier frame, repaired its own models when they broke, and cleanly separated a working demonstration from

the real-data test still owed. The losing side several times offered invalid or off-target counter-examples. But **on the substance the question stays open**: the duel showed the shadow mechanism is *sufficient* to manufacture the puzzle — it did not show that real brain data actually live in that regime.

Basis: real_compute (attached bundle) · **Protocol**: PASS (12 moves, clean arbiter verdict) · **Order-swap audit**: PASS (verdict stable when the side order was swapped) · **Arbiter confidence**: not assigned — substantive outcome marked unclear.

What this supports

- A purely lagged, shared, lopsided driver is **sufficient** to make static co-skewness predict cognition — reproducing the seed paper's puzzle with no genuine three-way term.
- Where that shadow exists, a directed lagged-connectivity model **absorbs** co-skewness almost entirely (increment ≈ 0.00 in the bundle).
- The signal genuinely requires a **shared** source: independent per-region drivers yield no predictive co-skewness.

What this does not support

- It does **not** show real fMRI co-skewness is such an artifact. The model is synthetic; it proves possibility, not actuality.
- It does **not** refute a genuine higher-order brain mechanism — no one could make a true instantaneous three-way term reproduce the puzzle stably, but absence here is not proof of absence in brains.
- It does **not** deliver the decisive test: a multi-lag / VAR directed baseline on real HCP data. That remains owed.

Where the losing side still has a point

Aramis's strongest move came at the end and is fair: the thesis claimed the increment falls *almost to zero*, yet on richer, more realistic baselines the residual was small but **run-to-run unstable** — sometimes near zero, sometimes not. "Almost zero" is a quantitative promise, and it was not met *stably*. Aramis proposed concrete thresholds for the real-data referee: on full HCP with a proper directed baseline, an increment below 0.02 would confirm the thesis, above 0.05 would refute it, and the band between is genuine uncertainty. That is the honest next experiment — and it is not one a synthetic world can settle.

Prior-work note

This release makes **no originality claim**. The seed paper established that co-skewness predicts cognition and that foundation models discard it; the duel's contribution is a stress-test of one mechanistic *explanation* for that signal, not a new empirical finding. Novelty status: **not an originality claim**. The real-data discriminator described above is, to the duel's knowledge, not yet reported.

Appendix · reproducibility

Numbers and figures come from the attached repro bundle. Script, data, hashes, environment and rerun instructions are recorded in **2026-06-06_compute_manifest.json**.

Model: 320 simulated subjects · 4 nodes · 300 timepoints · one common centered-exponential (skewed) driver spread to receiver nodes over a shared lag-1..3 kernel (weights 0.6 / 0.3 / 0.1); cognition modulates the directed coupling; a symmetric, cognition-independent nuisance loads every receiver. No genuine instantaneous three-way term by construction.

Estimator: ridge regression ($\alpha = 1.0$), 5-fold cross-validated R^2 ; mean $\pm$ std over 12 fixed seeds (0–11).

Environment: Python 3.12.3 · numpy 2.4.4 · linux-x86_64.

Headline values (common-source): co-skew alone $R^2 = 0.478$ (± 0.046) · static FC alone = 0.597 (± 0.054) · directed lagged FC alone = 0.908 (± 0.013) · co-skew increment over static FC = +0.027 (± 0.012) · over directed FC = -0.001 (± 0.002). Control (independent sources): co-skew alone = -0.031 (± 0.016).

script_sha256: sha256:f8d9ebffa2d98d9b3302203cd664c5a30ad06316f44203e960264d4adf8675a

2026-06-06_cc_data.json sha256: sha256:4efefeb98f085187ce1defccd9ac85dd137fbcbcf4c9fbbba83ed17717a2054be

Rerun: python3 2026-06-06_repro.py → writes 2026-06-06_cc_data.json

Errata / quarantine: none.

References

Marraffini, G., Mahuas, G., Borrell, T., Shevchenko, V., & Wassermann, D. (2026). The variance brain foundation models forgot: Third-order statistics predict cognition where billion-parameter models fail (arXiv:2606.04010). arXiv. <https://arxiv.org/abs/2606.04010>

Seed publication — the empirical finding the duel was built on.

Safaa, H., & Sabatini, B. L. (2026). Feature leakage and the identifiability of direct-dependency entropy models of neural activity (arXiv:2606.01661). arXiv. <https://arxiv.org/abs/2606.01661>

Supporting — the feature-leakage mechanism the thesis transfers onto the seed paper's empirics.

Citation style: APA 7. References ordered alphabetically by first author.

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